West Africa Variant Screener: Reproducible Evaluation of a Genomic Variant Prioritisation Platform

ClinVar Label Prediction, Serving Consistency and Data Coverage

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Abstract

The West Africa Variant Screener combines a browser interface, population-frequency annotations and a supervised classifier for exploring human genetic variants. This study evaluates the retained dataset and deployed random forest, asking how recorded performance changes under the application's actual inputs, removal of annotation-source metadata and stricter separation of training and test variants. The dataset contains 1,805 ClinVar-labelled variants in HBB, G6PD, HBA1 and HBA2. We reconstruct the recorded 1,444/361 training–test split, repeat fixed-protocol comparisons across five seeds, and examine grouped and gene-held-out splits. Baselines and ablations test the contributions of CADD, African/African American frequency annotations, a CADD-derived proxy and ClinVar review status. The saved forest reproduces historical AUC 0.9825 and 94.7% binary accuracy. Using serving-compatible review status gives AUC 0.9652 and 94.2% accuracy. The annotation-only forest reaches AUC 0.9669 on the primary split and 0.9312 when identical annotation groups are held out. Its AUC increment from AFR frequency is 0.0019 (paired bootstrap 95% interval [-0.0019, 0.0061]). The results describe internal discrimination between retained ClinVar label groups. They do not measure malaria infection, protection from malaria or patient-level diagnostic performance. The paper provides explicit split membership, predictions, uncertainty calculations and source checksums to support reproducible follow-up work.

Keywords: variant interpretation; ClinVar; population allele frequency; random forest; reproducibility; African ancestry.

1 Introduction

A variant's clinical meaning depends on the disease, inheritance pattern and evidence used to interpret it. A useful example is HBB: the sickle variant can contribute to an inherited disorder, while sickle-cell trait is associated with reduced risk of severe malaria. These are different outcomes. Large malaria association studies identify host loci associated with susceptibility, but those associations do not make a general pathogenicity classifier a malaria-resistance predictor [1].

The West Africa Variant Screener was developed to bring variant annotations and model output into a single research interface. It accepts a selected gene and numerical annotations, returns a classifier score and display tier, and provides gene and protein-structure reference views. The work is supervised by Dr. Klogo at CSIR Ghana.

The research question is whether the retained evidence supports the platform's performance claims under the conditions in which its model is used. The recorded evaluation uses variant-

specific ClinVar review status, whereas the application fixes this feature to one. Another field, named `conservation_phylop`, is a deterministic transform of CADD rather than an independently measured conservation score. Both details affect what a high evaluation score can establish.

We therefore examine recorded-model reproduction, serving consistency, feature ablations, baseline comparisons and split sensitivity. The contribution is an evaluation of an existing application and its data pipeline, with numerical artifacts that expose the comparison. No new biological assay or independently recruited clinical cohort is introduced.

2 Related work and interpretation context

ClinVar archives submitted classifications and their supporting evidence [2]. A retained ClinVar classification is the target in this study, rather than an independently established patient outcome. The source's review status describes the review process and agreement between submissions. Using that metadata as a predictor creates a different task from prediction using molecular annotations alone.

CADD integrates several annotations into a score for prioritising variants [4]. Its PHRED scale ranks variants; it is not a clinical probability. Population frequency provides another type of evidence. The gnomAD `afr` group refers to African/African American genetic ancestry in the source resource, rather than a Ghanaian or exclusively West African population sample [3]. These distinctions are retained throughout the evaluation.

Clinical interpretation combines population, computational, functional, segregation and other evidence in a disease-specific assessment [5]. ClinGen recommendations also emphasise calibration and the dependence between computational predictors [6]. The classifier studied here predicts a binary grouping of existing labels. Its score and three interface bands do not implement the full clinical evidence framework.

The model uses random forests [7] and synthetic minority oversampling (SMOTE) [8]. Both are established methods. The experiments compare a fixed forest configuration with simpler baselines and feature removals; they do not seek a new optimum on the historical test set.

3 Application and data pipeline

3.1 Scope and software architecture

The active application uses FastAPI, a saved scikit-learn model and a JavaScript browser interface. The model service loads a trusted artifact once, checks its feature order and class labels, and supports a strict mode that rejects missing or synthetic-placeholder models. For a prediction, it constructs six inputs in the recorded order, obtains the class-one score, and applies display rules. Figure 1 separates model-development data from interface inputs and reference views.

The interface accepts supplied annotations. Its demonstration presets do not retrieve a specific genomic variant. A manually selected protein residue changes the reference view and does not alter classifier input. The graphical feature bars are heuristic displays based on model importance and input-side rules; they are not per-variant SHAP attributions. Appendix C shows the existing interface as an illustration of these controls.

3.2 Retained dataset and label construction

The source parser selects exact target-gene symbols and GRCh38 records from a ClinVar variant-summary table. It maps benign, likely benign and their combined category to zero; pathogenic,

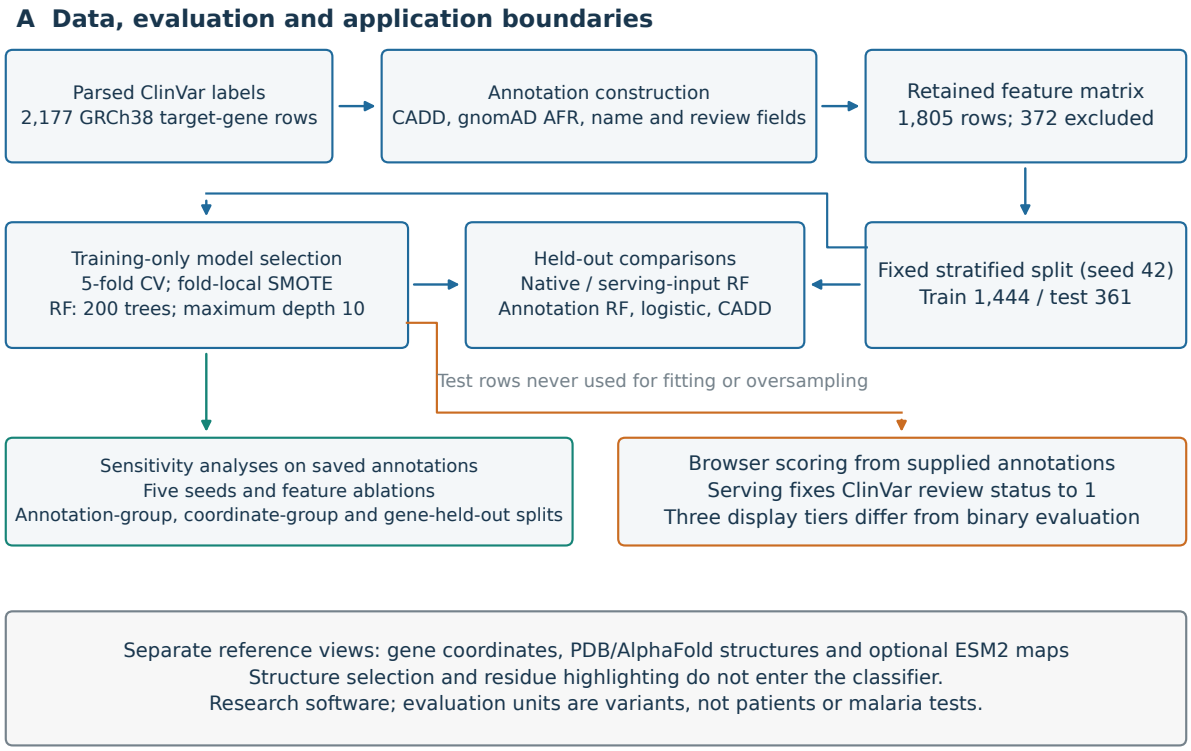


Figure 1: Data and application paths in the evaluated implementation. The retained matrix supplies model development and retrospective tests. Browser annotations supply serving inputs. Protein structures and optional language-model views are separate reference tools. No participant record enters the variant-level evaluation.

Gene	Parsed	Retained	Benign	Pathogenic
HBB	1249	1050	804	246
G6PD	555	523	313	210
HBA1	165	102	40	62
HBA2	208	130	42	88
DARC / ACKR1	0	0	0	0
Total	2,177	1,805	1,199	606

Table 1: Parsed and retained variant records by gene. The two label columns describe the retained matrix.

likely pathogenic and their combined category map to one. Other significance strings are excluded. Review status is encoded as zero to four stars, but the parser does not require a minimum star count. These operations leave 2,177 parsed records before feature-resolution exclusions.

Feature construction retains rows with nonmissing CADD and frequency fields. The frozen matrix contains 1,805 distinct variant identifiers. Its row order is preserved because reconstructing the historical split depends on it. The processed tables, code and original model are identified by SHA-256 checksums. The retained provenance sidecar marks the real-annotation branch, but does not record an immutable ClinVar release or raw-source checksum. Consequently, this study reproduces downstream evaluation of the frozen matrix; a new download may produce a different dataset.

Of the retained records, 1,199 are labelled benign/likely benign and 606 pathogenic/likely pathogenic. The target is the pooled ClinVar significance group, rather than a malaria-protection phenotype. DARC/ACKR1 has no retained record (Table 1).

Feature	Actual definition and interpretation
`allele_freq_afr`	Genome `afr` allele count divided by allele number, queried through `gnomad_r4` and rounded to six decimals. Missing usable population coverage can become zero.
`cadd_score`	PHRED-scaled CADD value. The copied configuration defaults to GRCh38-v1.7; retained rows do not carry per-query release metadata.
`gene_region`	A variant-name heuristic: 2 for names containing tokens such as `p.` or `c.`, 1 for UTR text, otherwise 0. It is not transcript-resolved annotation.
`conservation_phylop`	clip(CADD/40, −3, 3): a CADD-derived proxy, not independent phyloP.
`is_hbb`	One for HBB, zero for the other retained genes.
`review_stars`	ClinVar review-status mapping. Historical evaluation uses row values; the application fixes this input to one.

Table 2: The six recorded model inputs. Public source annotations, derived fields and label-source metadata have different evidential roles.

3.3 Feature definitions

The frequency client uses gnomAD genome population counts, not individual West African genomes. An unavailable or zero population denominator can be returned as zero frequency, and small positive frequencies can round to zero. A zero in the matrix therefore cannot be interpreted uniformly as observed absence. The 1000 Genomes population panel is downloadable by the project, but is not a separate training cohort for this classifier.

The proxy field duplicates information already present in CADD. Its retention in the historical model is reproduced faithfully; its removal is tested explicitly. The HBB flag also means that otherwise identical inputs for G6PD, HBA1, HBA2 and DARC yield identical classifier inputs. Selecting a gene name does not create a separately trained model for that gene.

3.4 Protein structures and optional reference tools

The structure registry uses experimental deoxyhaemoglobin PDB 2HHB for HBB (chain B) and the alpha-globin views (chain A) [9]. It uses PDB 2BH9, an experimental human G6PD deletion-variant structure, for the G6PD reference [10]. DARC/ACKR1 is configured with an AlphaFold-predicted reference [11]. These are reference structures, rather than measured structures of each entered variant. Residue numbering, sequence coverage and the identity of an experimental construct require checking before a specific residue-level interpretation.

The repository also implements optional ESM2-based mutational-context tools. Protein language models and zero-shot mutation scoring have separate methodological precedents [12, 13]. No cached ESM2 heatmap is part of the evaluated artifact set, and no ESM2 or structural feature enters the classifier. The prediction results below therefore do not measure the accuracy of either reference tool.

4 Experimental design

4.1 Frozen protocol and recorded-model reproduction

The primary reconstruction follows the recorded seed-42 stratified 80/20 split, producing 1,444 training and 361 test variants. The original model is evaluated only on that reconstructed historical holdout. The new experiment code saves row indices and variant identifiers for every split, together with predictions and source hashes. No test metric selects a checkpoint, feature set or hyperparameter.

Configuration	Inputs and purpose
Historical/native RF	Six recorded inputs; reproduces the historical feature definition.
Serving-input RF	The same fitted six-input forest, evaluated with review stars fixed to one.
No-review RF	Five inputs; removes the label-source review field during training and prediction.
Annotation RF	AFR frequency, CADD, region heuristic and HBB flag; removes review status and the duplicate proxy.
No-AFR RF	CADD, region heuristic and HBB flag; tests the incremental contribution of the frequency annotation.
Annotation logistic	Standardised four-input logistic regression, balanced class weights and 1,000-iteration limit; scaling fits training rows only.
CADD ranking	The supplied CADD score alone, evaluated for ranking without fitting or interpreting it as a probability.

Table 3: Prespecified baselines and ablations. Newly fitted alternatives are experimental comparisons and do not replace the deployed artifact.

The original forest configuration has 200 trees, maximum depth 10, minimum split size five and balanced class weights. Training uses SMOTE with its recorded defaults. Five-fold stratified cross-validation is restricted to training rows, with SMOTE fitted separately inside each fold. The full training split is then oversampled for the final fit. Standard SMOTE interpolates continuous and integer-coded columns alike; generated fractional values of encoded categories are audited rather than treated as biological observations.

The historical test has already been used in project development and documentation. Reconstructing it does not create an untouched external validation set. The repeated and stricter splits below are retrospective sensitivity analyses on the same frozen dataset.

4.2 Serving consistency and baseline comparisons

We evaluate the same original forest twice: once using the historical six-column test matrix, and once with `review_stars` fixed to one, matching the service. The remaining annotations are held unchanged. Metrics use unrounded model scores before the API's one-decimal percentage display. This isolates the measured effect of that serving-input difference for resolved dataset rows; it is not an evaluation of arbitrary manual inputs.

Fixed-protocol model comparisons use five split seeds: 42, 7, 19, 41 and 2026. Each forest ablation uses the same train/test rows as its corresponding full-feature run. The split seed also controls SMOTE and forest randomness. Hyperparameters remain fixed, including the default `max_features=sqrt`. Removing columns can therefore change the number of candidate features at each node as well as the available annotations. These ablations compare the specified algorithms rather than isolating a causal biological contribution.

4.3 Stricter separation and gene transfer

Two additional seed-42 group splits hold out approximately 20% of groups. One groups identical four-input annotation vectors; the other groups genomic chromosome–position pairs joined from the parsed table. Group counts rather than row counts determine the split, so the resulting test sizes and class proportions can differ. Forests and logistic models are refitted on each split's training rows. The original historical artifact is not reused against these new holdouts.

The feature-group test removes exact annotation-vector sharing across its partition. The position-group test removes shared genomic positions, but not every form of relatedness or identical annotation vector. Neither establishes independence from upstream predictor training or from neighbouring variants. Four leave-one-gene-out experiments additionally train the annotation

forest on three genes and test the fourth. They assess transfer across the retained gene set, rather than performance on unseen populations. DARC/ACKR1 has no retained training or test examples and receives no measured performance claim.

4.4 Metrics, intervals and feature diagnostics

For positive label one, we report ROC-AUC, average precision (AP), binary accuracy, sensitivity, specificity, F1 and Brier score. AP summarises precision–recall ranking; test prevalence supplies its reference level. Binary predictions use a 0.5 threshold. For probabilities p_i and labels y_i, the Brier score is

$$B = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2. \tag{1}$$

The interface instead assigns scores below 0.40 to its benign band, scores from 0.40 to below 0.75 to uncertain, and scores at least 0.75 to pathogenic. Tier changes are counted separately from binary prediction changes. These display bands do not reconstruct ClinVar's five clinical categories.

Primary-split uncertainty uses 2,000 stratified bootstrap resamples of held-out rows, with fixed bootstrap seed 20261108 and percentile 95% intervals. The same resampled indices are used for paired comparisons. These intervals are conditional on the fitted models, retained rows and row-resampling assumption. They do not include uncertainty from upstream labels, annotation resolution, model development or a target patient population. The five split results are reported individually with descriptive summaries; overlapping training and test partitions are not treated as independent clinical trials.

Reliability diagrams use five equal-width score bins to compare mean predicted scores with observed label fractions, with bin counts displayed. No calibration model is fitted on the test data. CADD is used as a ranking baseline and is excluded from probability-calibration comparisons. Gene and AFR-frequency subsets retain their own sample and class counts; AUC is undefined when only one label is present.

Grouped permutation diagnostics use 20 repetitions on the historical holdout. CADD and its derived proxy are permuted together to preserve their deterministic relationship. Frequency, region, HBB flag and review status are permuted separately. Drops in AUC describe dependence of the fixed model on these inputs under this perturbation, rather than causal or clinical importance.

4.5 Execution and provenance

The experiments ran on 8 November 2026 using an AMD Ryzen 5 PRO 2500U CPU and one forest job. The recorded environment is Python 3.14.8, scikit-learn 1.9.0, imbalanced-learn 0.14.2, NumPy 2.5.4, pandas 3.1.0 and SciPy 1.18.1. The complete numerical suite fitted 40 models: 25 fits across five row splits, six fits across two group splits, four gene-held-out fits and five cross-validation fits. It took 65.3 seconds including bootstrap, diagnostics and output writing. This is a recorded execution time, not a deployment capacity benchmark.

The source snapshot is project commit `13497ef`. No remote annotations were downloaded during these runs. Checksums identify the retained inputs and source files; immutable release metadata for the historical annotations is unavailable. Seventeen existing checks covering training-fold isolation, parsing, model loading and API behaviour also pass on the copied implementation. These are engineering checks rather than additional biological validation.

Model	AUC	AP	Accuracy	Sens.	Spec.	Brier
Native RF (6 inputs)	0.982	0.942	0.947	0.942	0.950	0.042
Serving-input RF	0.965	0.922	0.942	0.926	0.950	0.054
No-review RF	0.967	0.931	0.934	0.926	0.938	0.057
Annotation RF	0.967	0.932	0.934	0.926	0.938	0.057
No-AFR RF	0.965	0.933	0.920	0.926	0.917	0.063
Annotation logistic	0.957	0.943	0.936	0.926	0.942	0.061
CADD ranking	0.956	0.940	n/a	n/a	n/a	n/a

Table 4: Primary reconstructed holdout, 361 variants. The first two rows use the historically recorded forest definition; the other fitted models are retrospective comparisons. Binary metrics use 0.5. CADD has no probability or binary threshold in this analysis, so those entries are n/a.

5 Results

5.1 Coverage and retained-sample selection

Feature construction excludes 372 of 2,177 parsed records. It retains 95.0% of benign records and 66.2% of pathogenic records. Figure 2 shows the resulting coverage. The region heuristic codes 1,776 retained rows as exonic, 29 as intronic/other and 0 as UTR. AFR frequency is zero in 1,561 rows and positive in 244 rows.

The primary split has no repeated variant identifier or full genomic key across partitions. It does share 100 genomic positions, covering 102 test rows. Identical six-input vectors occur in both partitions for 51 groups and 58 test rows. Distinct variant identifiers therefore do not imply that all evaluated inputs are novel.

5.2 Historical reproduction and serving consistency

The fresh primary fit reproduces the saved forest's probabilities with maximum absolute difference 0.0 and no changed binary decisions. Training-only CV gives AUC 0.96997 ± 0.00525; the dispersion is fold standard deviation. The primary test contains 240 benign and 121 pathogenic records (positive prevalence 33.5%).

Historical AUC is 0.9825 (95% row-bootstrap interval [0.9684, 0.9936]), with binary accuracy 94.74%. Fixing review status to one gives AUC 0.9652 [0.9425, 0.9844] and accuracy 94.18%. The paired input change alters 6 binary decisions and 21 interface tiers (Figs. 3, 4 and 7).

5.3 Baselines and feature ablations

The annotation forest, which excludes review status and the duplicate proxy, has primary AUC 0.9669 [0.9432, 0.9856] and accuracy 93.35%. The logistic baseline gives AUC 0.9571, and raw CADD ranking gives 0.9559. Their AP values are 0.9434 and 0.9400, compared with 0.9416 for the native forest (Table 4). A ranking advantage on ROC-AUC does not imply an advantage on every metric.

Adding AFR frequency to the three-input forest changes primary AUC by +0.00186, with paired bootstrap interval [-0.0019, 0.0061]. The interval includes zero. This experiment does not establish an incremental frequency benefit. Figure 5 retains the results from every seed instead of selecting a favourable split.

Across five row splits, mean AUC is 0.9747 for native inputs, 0.9587 for serving inputs, 0.9540 for the annotation forest, 0.9567 for logistic regression and 0.9553 for CADD. Complete per-seed results appear in the appendix. These means describe sensitivity across overlapping internal

partitions.

5.4 Stricter splits and subgroup coverage

The annotation-group split has 1452 training and 353 test rows, including 109 positive test labels. Its annotation-forest AUC is 0.9312. Holding out chromosome–position groups gives 1447/358 training/test rows and AUC 0.9489. The changed populations and training sets preclude attributing the entire difference to one source of optimism (Fig. 6).

Leave-one-gene-out annotation-forest AUC is 0.9231 for HBB, 0.9255 for G6PD, 0.8675 for HBA1, 0.8703 for HBA2. These values do not establish equal accuracy across genes. Even on the native row holdout, HBA2 has accuracy 73.9% and specificity 44.4% among only 23 test variants (9 benign).

Only 6 primary test variants have AFR frequency above 0.01, and all are benign. The subgroup AUC is undefined. No West African patient-level accuracy follows from the pooled test score.

5.5 Model dependence and synthetic training rows

Joint permutation of CADD and its proxy lowers historical holdout AUC by 0.3694 on average. The mean drops are 0.0243 for review status and 0.0009 for AFR frequency. These perturbations identify dependence within the evaluated forest, rather than independent molecular evidence.

Primary SMOTE adds 474 synthetic rows to 1444 originals. Among those synthetic rows, 8 have fractional region codes, 52 fractional HBB flags and 67 fractional review-status codes. They are interpolated model inputs, not observed variants.

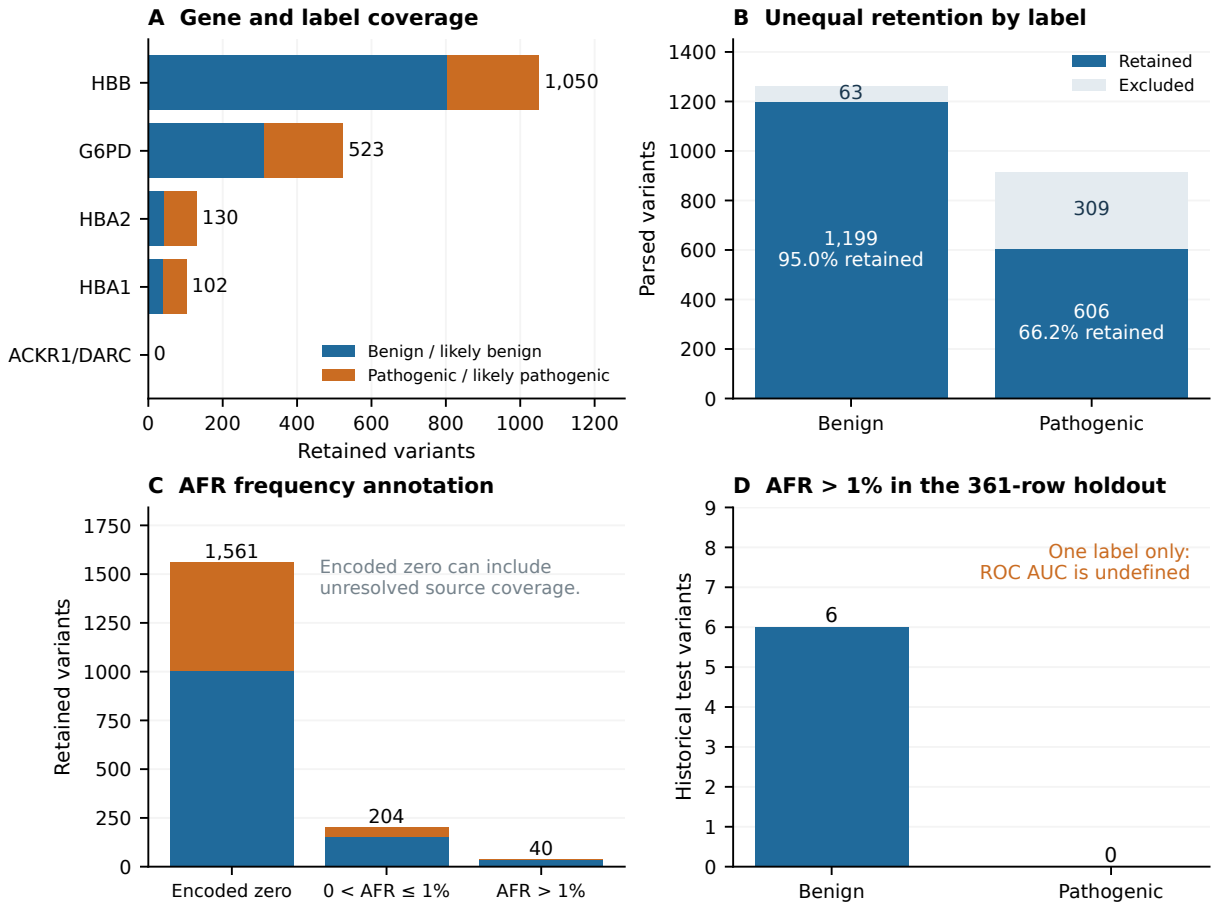


Figure 2: Coverage of the frozen dataset and feature-retention process. Counts refer to public variant records, not patients or application users. Zero frequency values include both biological and source-coverage possibilities; the retained matrix cannot separate them.

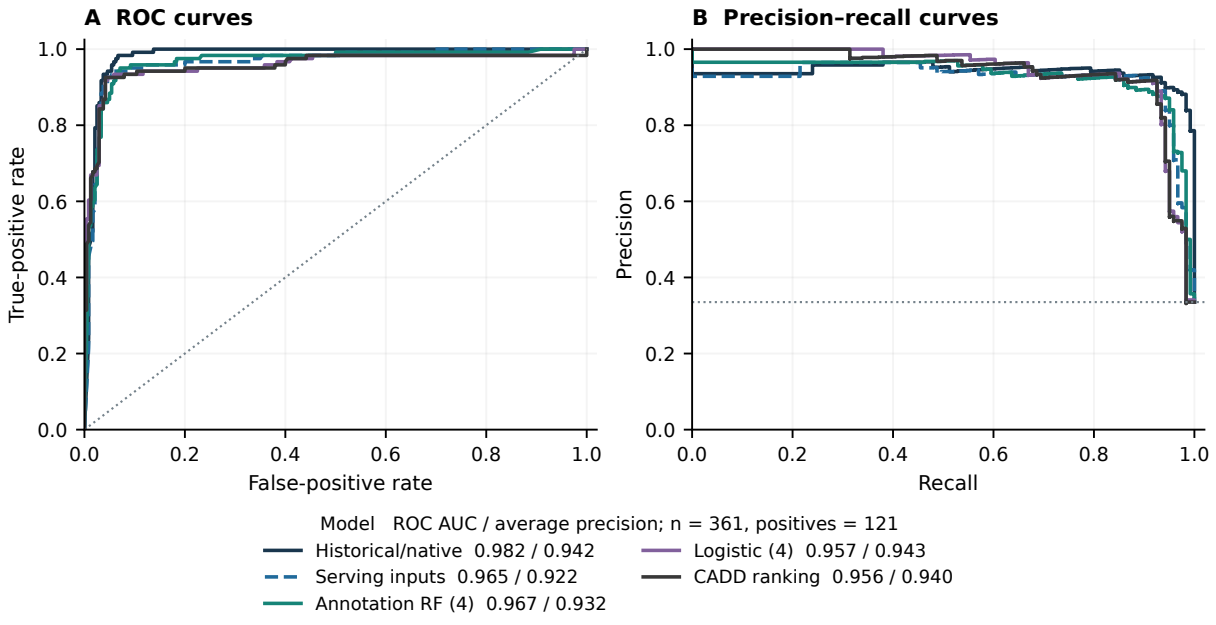


Figure 3: ROC and precision–recall curves on the reconstructed historical holdout. The serving-input curve uses the original forest with review stars fixed to one. Newly trained alternatives use the same partition. CADD is a ranking score. All curves use saved predictions; this is an internal retrospective comparison.

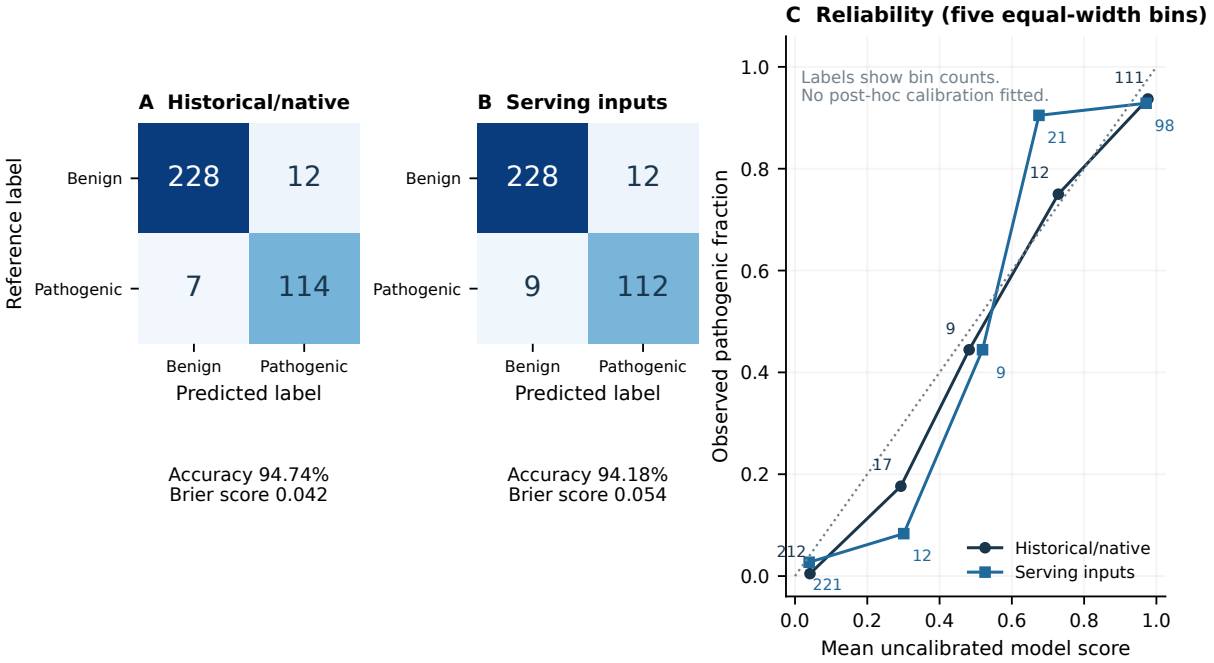


Figure 4: Binary confusion matrices and score reliability on the primary holdout. Binary classification uses 0.5, independently of the interface's 0.40 and 0.75 display thresholds. Reliability is measured against retained ClinVar labels and does not establish patient-risk calibration.

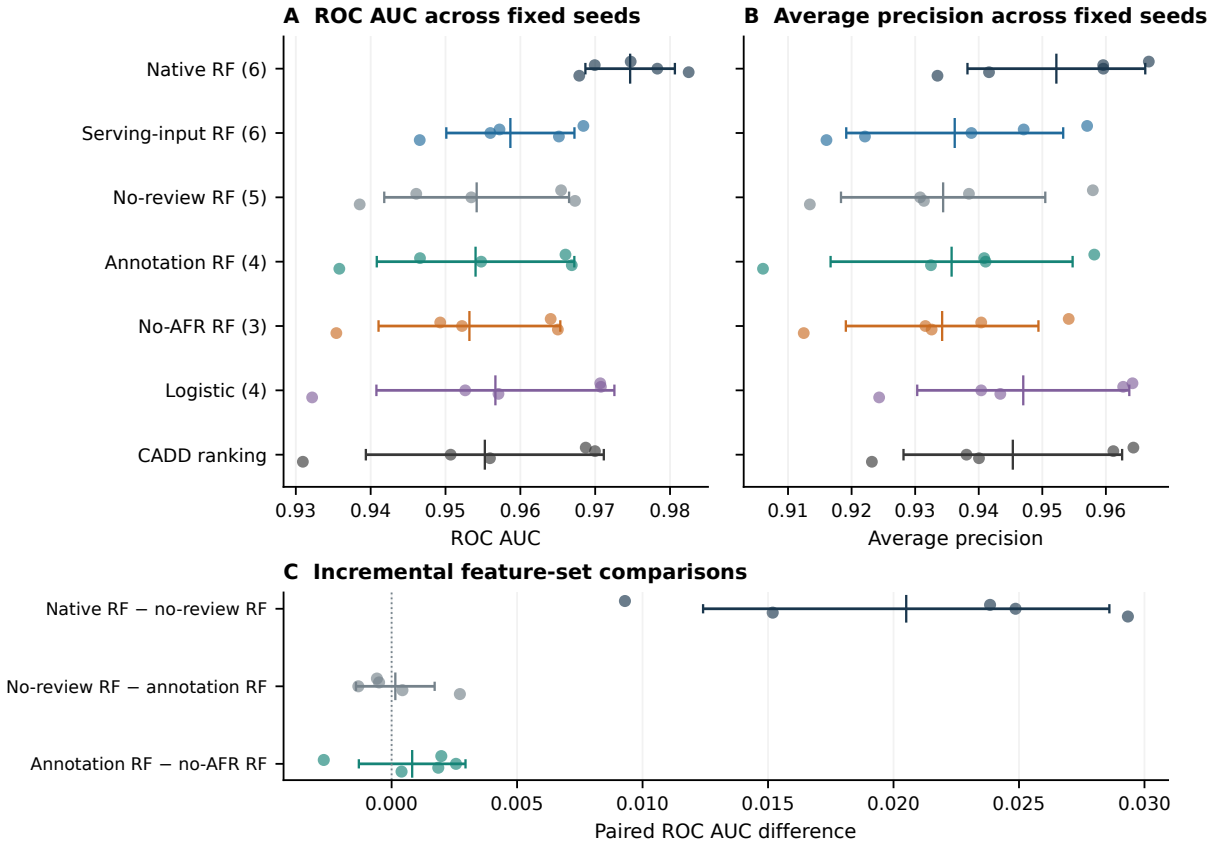


Figure 5: Fixed-protocol comparisons across five random splits, including paired AUC differences when review status, the duplicate proxy or AFR frequency is removed. Each point retains its split identity; bars show sample standard deviations, not confidence intervals. Models within a split share observations, and the splits reuse one dataset.

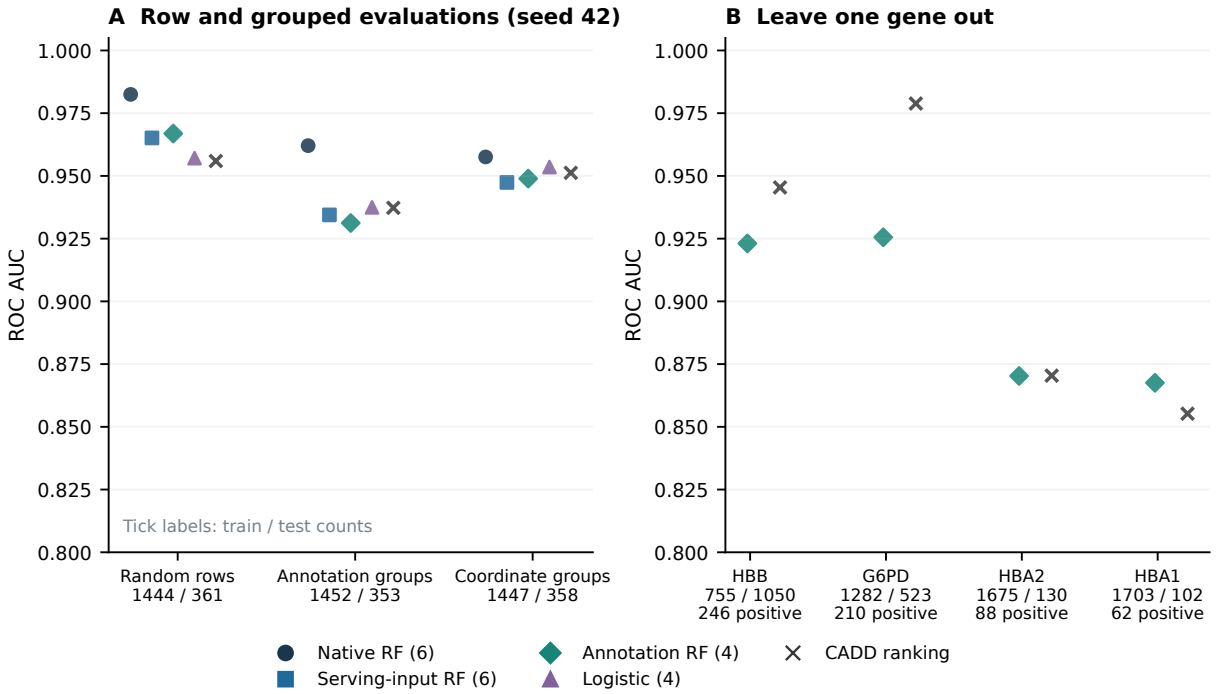


Figure 6: Stricter split and leave-one-gene-out results. Every model is refitted on the indicated training partition. Grouped holdout sizes and class proportions differ, so the plots should be read with their denominators. Gene transfer remains an internal test on four retained genes.

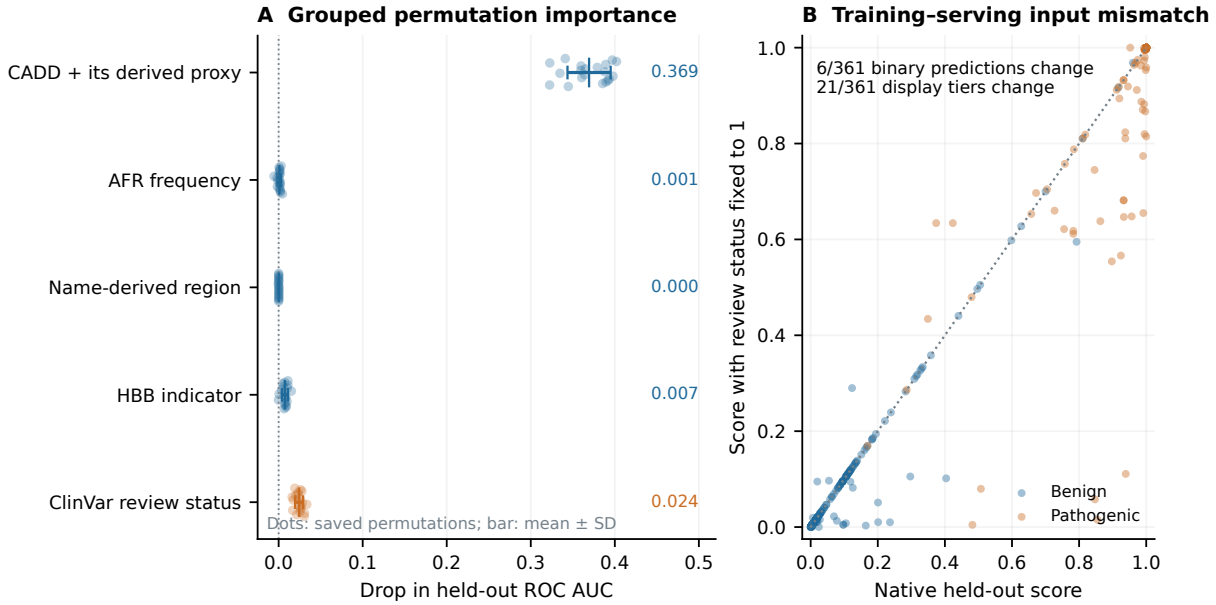


Figure 7: Grouped permutation diagnostics and the serving-input mismatch for the fixed historical forest. CADD and its proxy move together. Permutation variability describes 20 perturbations on this holdout. The second panel counts changed binary decisions and interface tiers when review status is fixed to one. The diagnostics are distinct from the interface's heuristic feature bars.

6 Discussion

6.1 What the performance estimates establish

The classifier separates the retained ClinVar label groups under the evaluated splits. Reproducing its historical score tests artifact consistency, while the serving-input analysis asks a different operational question: whether the historical feature definition matches the deployed service. The paired change in review status makes that distinction measurable. A high historical AUC alone does not establish the performance of the browser's actual input path.

The feature ablations further separate annotation-based prediction from dependence on source-review metadata. Removing the CADD-derived proxy removes no independent biological measurement, although it can change a forest's feature sampling and SMOTE geometry. The no-AFR comparison tests an incremental modelling contribution in this dataset. It does not measure the social value of population representation or establish accuracy in African patients.

The stricter splits expose different generalisation questions. A row split tests other retained variants under broadly shared feature patterns. Holding out exact annotation groups tests unseen combinations of those recorded inputs. Holding out genomic positions or entire genes changes the task again. Differences between these estimates therefore require interpretation alongside coverage and class balance, rather than selection of the most favourable score.

6.2 Population context and feature availability

The project's West African focus motivates examining allele-frequency evidence and its coverage. However, a gnomAD ancestry-group annotation is not a study cohort. The six high-frequency historical test variants cannot establish population-specific discrimination when all carry the same label. Even a larger frequency-defined subset would still differ from a geographically and clinically defined validation population.

Feature resolution also selects which variants reach the model. The loss of unresolved rows is unequal across label groups, and zero frequencies do not have a single interpretation. These issues affect the population of variants to which the measured results apply. Retaining explicit missingness states, annotation versions and transcript-aware consequences would make a future comparison easier to interpret.

6.3 Application use and reference views

Project documentation records application use by 500 people. This is a platform-use count, separate from the 1,805 model-development variants and 361 historical test variants. Participant-level records, recruitment details and numerical usability or clinical outcomes are not analysed here.

The interface can make annotations and model behaviour easier to inspect, but a displayed structure is not evidence that a submitted variant changes that structure. The experimental G6PD construct and the configured AlphaFold view also have their own identity and coverage limits. Keeping these views separate from classifier inputs avoids attributing the forest's performance to information it never received.

7 Limitations and next experiments

The study uses a frozen, selected ClinVar-derived matrix rather than a prospectively collected patient cohort. The historical test was already part of development, and the new split analyses

remain retrospective. Upstream predictors and ClinVar can share knowledge or variants, which the present row, group and gene splits cannot fully audit. Conditional bootstrap intervals do not resolve those sources of uncertainty.

The feature definitions constrain inference. Review status is tied to the label source and differs at serving. The conservation field duplicates CADD. Gene region is inferred from text, and the gene encoding distinguishes only HBB from the remaining options. SMOTE interpolates encoded categories. DARC/ACKR1 has no labelled examples, and no UTR-coded row is present. These are properties of the implementation evaluated here, rather than independent validation of its selectable input ranges.

The next development step is a versioned annotation pipeline that preserves missingness, transcript consequences and source release dates, followed by a serving-compatible model evaluated on data withheld from development. Comparisons should include an independent source or temporal holdout and a suitably defined target population. Prospective clinical or usability claims would require their own study designs and outcome denominators. Optional protein-language-model and structure tools also need separate benchmarks before receiving accuracy claims.

8 Conclusion

The West Africa Variant Screener combines annotation-based scoring with a browser research interface. The frozen historical model reproduces AUC 0.9825; matching the service's fixed review-status input changes it to 0.9652. An annotation-only forest gives AUC 0.9669 on the primary split and 0.9312 under annotation-group holdout. Simple baselines remain informative, and the measured AFR-frequency increment is not established by the paired interval. The study provides an auditable internal evaluation with explicit coverage and serving limits. Population-specific, clinical and protein-reference accuracy require separate evidence.

Author contribution and supervision

I developed the Variant Screener under the supervision of Dr. Klogo at CSIR Ghana.

Data, code and availability

Application source: West Africa Variant Screener on GitHub. The analysis uses a fixed source snapshot, processed public variant-level tables and the original model. Split membership, predictions, metrics and source checksums are retained with the study records. Appendix A describes these records and their reproducibility limits. Participant-level outcomes are not part of this analysis.

Preparation statement

AI assistance was used for code inspection, experiment scripting, figure preparation and prose editing. All numerical figures were generated from the retained variant data and saved experimental outputs.

References

[1] Malaria Genomic Epidemiology Network. Insights into malaria susceptibility using genome-wide data on 17,000 individuals from Africa, Asia and Oceania. *Nature Communications*. 2019;10:5732. https://doi.org/10.1038/s41467-019-13480-z.

[2] Landrum MJ, Chitipiralla S, Kaur K, et al. ClinVar: updates to support classifications of both germline and somatic variants. *Nucleic Acids Research*. 2025;53(D1):D1313–D1321. https://doi.org/10.1093/nar/gkae1090. Review-status documentation: https://www.ncbi.nlm.nih.gov/clinvar/docs/review_status/ (accessed 8 November 2026).

[3] Chao K, gnomAD Production Team. gnomAD v4.0; Genetic Ancestry. Official release and ancestry documentation, 1 November 2023. https://gnomad.broadinstitute.org/news/2023-11-gnomad-v4-0/; https://gnomad.broadinstitute.org/news/2023-11-genetic-ancestry (accessed 8 November 2026).

[4] Schubach M, Maass T, Nazaretyan L, Röner S, Kircher M. CADD v1.7: using protein language models, regulatory CNNs and other nucleotide-level scores to improve genome-wide variant predictions. *Nucleic Acids Research*. 2024;52(D1):D1143–D1154. https://doi.org/10.1093/nar/gkad989.

[5] Richards S, Aziz N, Bale S, et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in Medicine*. 2015;17(5):405–424. https://doi.org/10.1038/gim.2015.30.

[6] Pejaver V, Byrne AB, Feng BJ, et al.; ClinGen Sequence Variant Interpretation Working Group. Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria. *American Journal of Human Genetics*. 2022;109(12):2163–2177. https://doi.org/10.1016/j.ajhg.2022.10.013.

[7] Breiman L. Random forests. *Machine Learning*. 2001;45:5–32. https://doi.org/10.1023/A:1010933404324.

[8] Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP. SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*. 2002;16:321–357. https://doi.org/10.1613/jair.953.

[9] Fermi G, Perutz MF, Shaanan B, Fourme R. The crystal structure of human deoxyhaemoglobin at 1.74 Å resolution. *Journal of Molecular Biology*. 1984;175:159–174. https://doi.org/10.1016/0022-2836(84)90472-8. Structure record: RCSB PDB 2HHB, https://doi.org/10.2210/pdb2HHB/pdb (accessed 8 November 2026).

[10] Kotaka M, Gover S, Vandeputte-Rutten L, Au SWN, Lam VMS, Adams MJ. Structural studies of glucose-6-phosphate and $NADP^+$ binding to human glucose-6-phosphate dehydrogenase. *Acta Crystallographica Section D: Biological Crystallography*. 2005;61(Pt 5):495–504. https://doi.org/10.1107/S0907444905002350. Structure record: RCSB PDB 2BH9, https://doi.org/10.2210/pdb2BH9/pdb (accessed 8 November 2026).

[11] Varadi M, Bertoni D, Magana P, et al. AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences. *Nucleic Acids Research*. 2024;52(D1):D368–D375. https://doi.org/10.1093/nar/gkad1011. Prediction-confidence and limitation documentation: https://alphafold.ebi.ac.uk/faq (accessed 8 November 2026).

[12] Lin Z, Akin H, Rao R, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*. 2023;379(6637):1123–1130. https://doi.org/10.1126/science.ade2574.

[13] Meier J, Rao R, Verkuil R, Liu J, Sercu T, Rives A. Language models enable zero-shot prediction of the effects of mutations on protein function. *Advances in Neural Information Processing Systems*. 2021;34:29287–29303. https://proceedings.neurips.cc/paper/2021/hash/f51338d736f95dd42427296047067694-Abstract.html.

A Reproducibility records

The analysis preserves the processed matrix in its original row order and records the source snapshot, model configuration and software environment. Split files identify the training and test variants for each comparison. Prediction files retain each held-out label, score and binary decision, allowing the reported metrics to be recalculated without fitting the models again.

Record	Contents
Source and input checksums	Application source snapshot, retained variant tables and original model.
Experiment protocol	Model, split, bootstrap and permutation settings.
Execution record	Software environment, CPU, timing and fit counts.
Split membership	Explicit training, test and cross-validation assignments.
Predictions	Every held-out label, score and binary decision.
Metrics	Fifty-five model/evaluation rows, including ranking-only baselines.
Bootstrap results	All 2,000 paired stratified resamples.
Analysis models	Five fitted primary models with feature and split metadata.
Figures	Seven scientific figures and the separate interface image.

Table 5: Records retained for the numerical assessment.

The fixed matrix supports reproduction of the downstream analysis. Missing historical release metadata prevents exact reconstruction of its upstream annotation downloads. The study records contain variant-level data and do not include a 500-participant outcome dataset.

B Complete sensitivity results

Table 6: Every row-split result. Each split has 361 test variants, including 121 positive labels. Seeds reuse the same dataset.

Seed	Model	AUC	AP	Accuracy	Brier
42	Native RF (6 inputs)	0.9825	0.9416	0.9474	0.0421
42	Serving-input RF	0.9652	0.9221	0.9418	0.0543
42	No-review RF	0.9673	0.9314	0.9335	0.0568
42	Annotation RF	0.9669	0.9325	0.9335	0.0566
42	No-AFR RF	0.9650	0.9326	0.9197	0.0632
42	Annotation logistic	0.9571	0.9434	0.9363	0.0612
42	CADD ranking	0.9559	0.9400	n/a	n/a
7	Native RF (6 inputs)	0.9747	0.9668	0.9501	0.0404
7	Serving-input RF	0.9684	0.9570	0.9418	0.0484
7	No-review RF	0.9654	0.9579	0.9474	0.0498
7	Annotation RF	0.9660	0.9582	0.9418	0.0499
7	No-AFR RF	0.9640	0.9542	0.9446	0.0530
7	Annotation logistic	0.9707	0.9642	0.9418	0.0500
7	CADD ranking	0.9687	0.9643	n/a	n/a
19	Native RF (6 inputs)	0.9699	0.9596	0.9474	0.0444
19	Serving-input RF	0.9572	0.9471	0.9335	0.0547
19	No-review RF	0.9461	0.9385	0.9335	0.0589
19	Annotation RF	0.9466	0.9409	0.9418	0.0570
19	No-AFR RF	0.9493	0.9404	0.9280	0.0614
19	Annotation logistic	0.9708	0.9627	0.9474	0.0505
19	CADD ranking	0.9700	0.9612	n/a	n/a
41	Native RF (6 inputs)	0.9783	0.9596	0.9446	0.0494
41	Serving-input RF	0.9560	0.9388	0.9280	0.0619
41	No-review RF	0.9534	0.9308	0.9280	0.0650

Seed	Model	AUC	AP	Accuracy	Brier
41	Annotation RF	0.9548	0.9411	0.9307	0.0638
41	No-AFR RF	0.9522	0.9316	0.9224	0.0683
41	Annotation logistic	0.9526	0.9404	0.9307	0.0643
41	CADD ranking	0.9507	0.9381	n/a	n/a
2026	Native RF (6 inputs)	0.9679	0.9335	0.9418	0.0562
2026	Serving-input RF	0.9466	0.9160	0.9224	0.0672
2026	No-review RF	0.9385	0.9134	0.9280	0.0708
2026	Annotation RF	0.9358	0.9061	0.9252	0.0722
2026	No-AFR RF	0.9354	0.9125	0.9141	0.0734
2026	Annotation logistic	0.9322	0.9243	0.9307	0.0650
2026	CADD ranking	0.9309	0.9232	n/a	n/a

Split	Model	Test n	Positive	AUC	Accuracy
Row	Native RF (6 inputs)	361	121	0.9825	0.9474
Row	Serving-input RF	361	121	0.9652	0.9418
Row	Annotation RF	361	121	0.9669	0.9335
Row	Annotation logistic	361	121	0.9571	0.9363
Row	CADD ranking	361	121	0.9559	n/a
Annotation groups	Native RF (6 inputs)	353	109	0.9621	0.9235
Annotation groups	Serving-input RF	353	109	0.9345	0.9207
Annotation groups	Annotation RF	353	109	0.9312	0.9093
Annotation groups	Annotation logistic	353	109	0.9375	0.9292
Annotation groups	CADD ranking	353	109	0.9373	n/a
Position groups	Native RF (6 inputs)	358	114	0.9576	0.9106
Position groups	Serving-input RF	358	114	0.9474	0.8966
Position groups	Annotation RF	358	114	0.9489	0.8939
Position groups	Annotation logistic	358	114	0.9535	0.9022
Position groups	CADD ranking	358	114	0.9512	n/a

Table 7: Seed-42 split stress tests, with newly fitted models for each partition.

Held-out gene	Model	Train	Test	Positive	AUC	Accuracy
HBB	Annotation RF	755	1050	246	0.9231	0.9324
HBB	CADD ranking	755	1050	246	0.9454	n/a
G6PD	Annotation RF	1282	523	210	0.9255	0.7686
G6PD	CADD ranking	1282	523	210	0.9789	n/a
HBA1	Annotation RF	1703	102	62	0.8675	0.7451
HBA1	CADD ranking	1703	102	62	0.8552	n/a
HBA2	Annotation RF	1675	130	88	0.8703	0.7769
HBA2	CADD ranking	1675	130	88	0.8704	n/a

Table 8: Leave-one-gene-out tests. The HBB flag is constant in training when HBB is held out.

Gene	Inputs	Test	Positive	AUC	Accuracy	Specificity
HBB	Native	223	60	0.9823	0.9507	0.9632
HBB	Serving	223	60	0.9678	0.9462	0.9632
G6PD	Native	102	37	0.9992	0.9902	0.9846
G6PD	Serving	102	37	0.9983	0.9902	0.9846
HBA1	Native	13	10	1.0000	0.9231	1.0000
HBA1	Serving	13	10	0.9000	0.8462	1.0000
HBA2	Native	23	14	0.9206	0.7391	0.4444
HBA2	Serving	23	14	0.8889	0.7391	0.4444

Table 9: Gene subgroups within the original row holdout. Small counts limit each estimate.

C Interface documentation

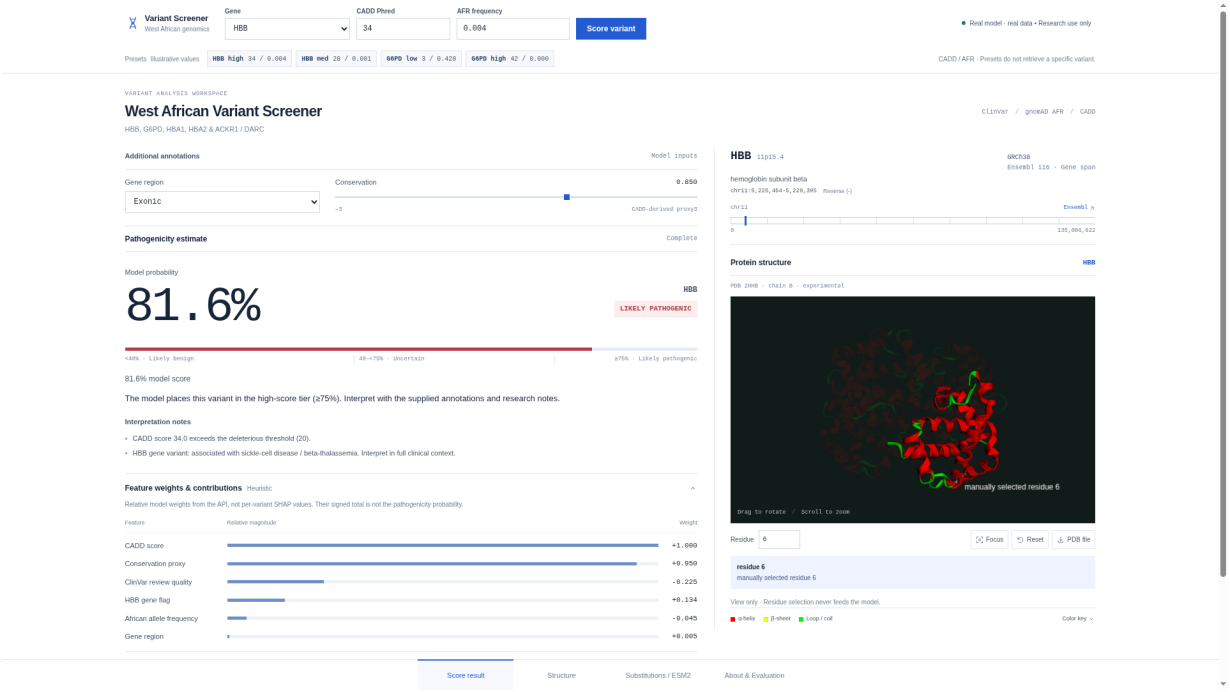


Figure 8: Existing Variant Screener workspace. The displayed HBB preset is an illustrative annotation example, not a retrieved or validated HbS variant. The 81.6% displayed score is separate from the measured test metrics. The structure is experimental PDB 2HHB, chain B; residue selection is manual and does not enter the model. The feature bars are labelled heuristic.

The browser view and model service are included in the source snapshot. Optional structure retrieval and ESM2 views are outside the classifier experiments, and the screenshot is interface documentation rather than a scientific prediction plot.